

$$\sigma_{22}^2 (y_1 - \mu_1)^2 - 2 \sigma_{12} \sigma_{22} (y_1 - \mu_1) (y_2 - \mu_2) + \sigma_{12}^2 (y_2 - \mu_2)^2 + \sigma_{21}^2 (y_1 - \mu_1)^2 - 2 \sigma_{11} \sigma_{21} (y_1 - \mu_1) (y_2 - \mu_2) + \sigma_{11}^2 (y_2 - \mu_2)^2, \quad (22)$$

so

$$\begin{aligned} & (x_1^2 + x_2^2) (\sigma_{11} \sigma_{22} - \sigma_{12} \sigma_{21})^2 \\ &= (y_1 - \mu_1)^2 (\sigma_{21}^2 + \sigma_{22}^2) - 2 (y_1 - \mu_1) (y_2 - \mu_2) (\sigma_{11} \sigma_{21} + \sigma_{12} \sigma_{22}) + (y_2 - \mu_2)^2 (\sigma_{11}^2 + \sigma_{12}^2) \\ &= \sigma_{22}^2 (y_1 - \mu_1)^2 - 2 (y_1 - \mu_1) (y_2 - \mu_2) (\rho \sigma_1 \sigma_2) + \sigma_{11}^2 (y_2 - \mu_2)^2 \\ &= \sigma_1^2 \sigma_2^2 \left[\frac{(y_1 - \mu_1)^2}{\sigma_1^2} - \frac{2 \rho (y_1 - \mu_1) (y_2 - \mu_2)}{\sigma_1 \sigma_2} + \frac{(y_2 - \mu_2)^2}{\sigma_2^2} \right]. \end{aligned} \quad (23)$$

Now, the denominator of (21) is

$$\begin{aligned} & \sigma_{11}^2 \sigma_{21}^2 + \sigma_{11}^2 \sigma_{22}^2 + \sigma_{12}^2 \sigma_{21}^2 + \sigma_{12}^2 \sigma_{22}^2 - \sigma_{11}^2 \sigma_{21}^2 \\ & - 2 \sigma_{11} \sigma_{12} \sigma_{21} \sigma_{22} - \sigma_{12}^2 \sigma_{22}^2 = (\sigma_{11} \sigma_{22} - \sigma_{12} \sigma_{21})^2, \end{aligned} \quad (24)$$

so

$$\frac{1}{1 - \rho^2} = 1 - \frac{1}{\frac{\sigma_1^2 \sigma_2^2}{\sigma_{11}^2 \sigma_{22}^2}} \quad (25)$$

$$= \frac{\sigma_1^2 \sigma_2^2}{\sigma_{11}^2 \sigma_{22}^2 - \sigma_{12}^2} \quad (26)$$

$$= \frac{\sigma_1^2 \sigma_2^2}{(\sigma_{11}^2 + \sigma_{12}^2) (\sigma_{21}^2 + \sigma_{22}^2) - (\sigma_{11} \sigma_{21} + \sigma_{12} \sigma_{22})^2}. \quad (27)$$

can be written simply as

$$\frac{1}{1 - \rho^2} = \frac{\sigma_1^2 \sigma_2^2}{(\sigma_{11} \sigma_{22} - \sigma_{12} \sigma_{21})^2}, \quad (28)$$

and

$$x_1^2 + x_2^2 = \frac{1}{1 - \rho^2} \left[\frac{(y_1 - \mu_1)^2}{\sigma_1^2} - \frac{2 \rho (y_1 - \mu_1) (y_2 - \mu_2)}{\sigma_1 \sigma_2} + \frac{(y_2 - \mu_2)^2}{\sigma_2^2} \right]. \quad (29)$$

Solving for x_1 and x_2 and defining

$$\rho' = \frac{\sigma_1 \sigma_2 \sqrt{1 - \rho^2}}{\sigma_{11} \sigma_{22} - \sigma_{12} \sigma_{21}} \quad (30)$$

gives

$$x_1 = \frac{\sigma_{22} (y_1 - \mu_1) - \sigma_{12} (y_2 - \mu_2)}{\rho'} \quad (31)$$

$$x_2 = \frac{-\sigma_{21} (y_1 - \mu_1) + \sigma_{11} (y_2 - \mu_2)}{\rho'}. \quad (32)$$

But the Jacobian is

$$J \begin{pmatrix} x_1, x_2 \\ y_1, y_2 \end{pmatrix} = \begin{vmatrix} \frac{\partial x_1}{\partial y_1} & \frac{\partial x_1}{\partial y_2} \\ \frac{\partial x_2}{\partial y_1} & \frac{\partial x_2}{\partial y_2} \end{vmatrix} = \begin{vmatrix} \frac{\sigma_{22}}{\rho'} & -\frac{\sigma_{12}}{\rho'} \\ -\frac{\sigma_{21}}{\rho'} & \frac{\sigma_{11}}{\rho'} \end{vmatrix} \quad (33)$$

$$= \frac{1}{\rho'^2} (\sigma_{11} \sigma_{22} - \sigma_{12} \sigma_{21}) \quad (34)$$

$$= \frac{1}{\rho'} = \frac{1}{\sigma_1 \sigma_2 \sqrt{1 - \rho^2}}, \quad (35)$$

so

$$d x_1 d x_2 = \frac{d y_1 d y_2}{\sigma_1 \sigma_2 \sqrt{1 - \rho^2}} \quad (36)$$

and

$$\frac{1}{2 \pi} e^{-\frac{1}{2}(x_1^2 + x_2^2)} d x_1 d x_2 = \frac{1}{2 \pi \sigma_1 \sigma_2 \sqrt{1 - \rho^2}} \exp \left[-\frac{z}{2(1 - \rho^2)} \right] d y_1 d y_2, \quad (37)$$

where

$$z = \frac{(y_1 - \mu_1)^2}{\sigma_1^2} - \frac{2 \rho (y_1 - \mu_1) (y_2 - \mu_2)}{\sigma_1 \sigma_2} + \frac{(y_2 - \mu_2)^2}{\sigma_2^2}. \quad (38)$$

Q.E.D.

The characteristic function of the bivariate normal distribution is given by

$$\phi(t_1, t_2) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{i(t_1 x_1 + t_2 x_2)} P(x_1, x_2) d x_1 d x_2 \quad (39)$$

$$= N \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{i(t_1 x_1 + t_2 x_2)} \exp \left[-\frac{z}{2(1 - \rho^2)} \right] d x_1 d x_2, \quad (40)$$

where

$$z = \left[\frac{(x_1 - \mu_1)^2}{\sigma_1^2} - \frac{2 \rho (x_1 - \mu_1) (x_2 - \mu_2)}{\sigma_1 \sigma_2} + \frac{(x_2 - \mu_2)^2}{\sigma_2^2} \right] \quad (41)$$

and

$$N = \frac{1}{2 \pi \sigma_1 \sigma_2 \sqrt{1 - \rho^2}}. \quad (42)$$

Now let

$$u \equiv x_1 - \mu_1 \quad (43)$$

$$w \equiv x_2 - \mu_2. \quad (44)$$

Then

$$\phi(t_1, t_2) = N' \int_{-\infty}^{\infty} \left(e^{i t_2 w} \exp \left[-\frac{1}{2(1 - \rho^2)} \frac{w^2}{\sigma_2^2} \right] \right) \int_{-\infty}^{\infty} e^{i t_1 u} d u d w, \quad (45)$$

where

$$v \equiv -\frac{1}{2(1 - \rho^2)} \frac{1}{\sigma_1^2} \left[u^2 - \frac{2 \rho \sigma_1 w}{\sigma_2} u \right] \quad (46)$$

$$N' = \frac{e^{i(t_1 \mu_1 + t_2 \mu_2)}}{2 \pi \sigma_1 \sigma_2 \sqrt{1 - \rho^2}}. \quad (47)$$

Complete the square in the inner integral

$$\begin{aligned} & \int_{-\infty}^{\infty} \exp \left[-\frac{1}{2(1 - \rho^2)} \frac{1}{\sigma_1^2} \left[u^2 - \frac{2 \rho \sigma_1 w}{\sigma_2} u \right] \right] e^{i t_1 u} d u \\ &= \int_{-\infty}^{\infty} \exp \left[-\frac{1}{2 \sigma_1^2 (1 - \rho^2)} \left[u - \frac{\rho_1 \sigma_1 w}{\sigma_2} \right]^2 \right] \left\{ \frac{1}{2 \sigma_1^2 (1 - \rho^2)} \left(\frac{\rho_1 \sigma_1 w}{\sigma_2} \right)^2 \right\} e^{i t_1 u} d u. \end{aligned} \quad (48)$$

Rearranging to bring the exponential depending on w outside the inner integral, letting

$$v \equiv u - \rho \frac{\sigma_1 w}{\sigma_2}, \quad (49)$$

and writing

$$e^{i t_1 u} = \cos(t_1 u) + i \sin(t_1 u) \quad (50)$$

gives

$$\begin{aligned} \phi(t_1, t_2) = & N' \int_{-\infty}^{\infty} e^{i t_2 w} \exp \left[-\frac{1}{2 \sigma_2^2 (1 - \rho^2)} w^2 \right] \exp \left[\frac{\rho^2}{2 \sigma_2^2 (1 - \rho^2)} w^2 \right] \int_{-\infty}^{\infty} \exp \left[-\frac{1}{2 \sigma_1^2 (1 - \rho^2)} v^2 \right] \left\{ \cos \left[t_1 \left(v + \frac{\rho \sigma_1 w}{\sigma_2} \right) \right] + i \sin \left[t_1 \left(v + \frac{\rho \sigma_1 w}{\sigma_2} \right) \right] \right\} d v d w. \end{aligned} \quad (51)$$

Expanding the term in braces gives

$$\begin{aligned} & \left[\cos(t_1 \nu) \cos\left(\frac{\rho \sigma_1 w t_1}{\sigma_2}\right) - \sin(t_1 \nu) \sin\left(\frac{\rho \sigma_1 w}{\sigma_2 t_1}\right) \right] + i \left[\sin(t_1 \nu) \cos\left(\frac{\rho \sigma_1 w}{\sigma_2 t_1}\right) + \cos(t_1 \nu) \sin\left(\frac{\rho \sigma_1 w t_1}{\sigma_2}\right) \right] \\ &= \left[\cos\left(\frac{\rho \sigma_1 w t_1}{\sigma_2}\right) + i \sin\left(\frac{\rho \sigma_1 w t_1}{\sigma_2}\right) \right] [\cos(t_1 \nu) + i \sin(t_1 \nu)] = \exp\left(\frac{i \rho \sigma_1 w}{\sigma_2} t_1\right) [\cos(t_1 \nu) + i \sin(t_1 \nu)]. \end{aligned} \tag{52}$$

But $e^{-a x^2} \sin(b x)$ is odd, so the integral over the sine term vanishes, and we are left with

$$\begin{aligned} \phi(t_1, t_2) &= N' \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{i t_2 w} \exp\left[-\frac{w^2}{2 \sigma_2^2}\right] \exp\left[-\frac{\rho^2 w^2}{2 \sigma_2^2 (1-\rho^2)}\right] \exp\left[\frac{i \rho \sigma_1 w t_1}{\sigma_2}\right] d w \int_{-\infty}^{\infty} \exp\left[-\frac{v^2}{2 \sigma_1^2 (1-\rho^2)}\right] \cos(t_1 \nu) d \nu \\ &= N' \int_{-\infty}^{\infty} \exp\left[i w \left(t_2 + t_1 \left(\frac{\rho \sigma_1}{\sigma_2}\right)\right)\right] \exp\left[-\frac{w^2}{2 \sigma_2^2}\right] d w \int_{-\infty}^{\infty} \exp\left[-\frac{v^2}{2 \sigma_1^2 (1-\rho^2)}\right] \cos(t_1 \nu) d \nu. \end{aligned} \tag{53}$$

Now evaluate the Gaussian integral

$$\int_{-\infty}^{\infty} e^{i k x} e^{-a x^2} d x = \int_{-\infty}^{\infty} e^{-a x^2} \cos(k x) d x \tag{54}$$

$$= \sqrt{\frac{\pi}{a}} e^{-k^2/4 a} \tag{55}$$

to obtain the explicit form of the characteristic function,

$$\begin{aligned} \phi(t_1, t_2) &= \frac{e^{i(t_1 \mu_1 + t_2 \mu_2)}}{2 \pi \sigma_1 \sigma_2 \sqrt{1-\rho^2}} \left\{ \sigma_2 \sqrt{2 \pi} \exp\left[-\frac{1}{4} \left(t_2 + \rho \frac{\sigma_1}{\sigma_2} t_1\right)^2 2 \sigma_2^2\right] \left\{ \sigma_1 \sqrt{2 \pi (1-\rho^2)} \exp\left[-\frac{1}{4} t_1^2 2 \sigma_1^2 (1-\rho^2)\right] \right\} \right. \\ &= e^{i(t_1 \mu_1 + t_2 \mu_2)} \exp\left\{-\frac{1}{2} \left[t_2^2 \sigma_2^2 + 2 \rho \sigma_1 \sigma_2 t_1 t_2 + \rho^2 \sigma_1^2 t_1^2 + (1-\rho^2) \sigma_1^2 t_1^2\right]\right\} \\ &= \exp\left[i(t_1 \mu_1 + t_2 \mu_2) - \frac{1}{2} (\sigma_1^2 t_1^2 + 2 \rho \sigma_1 \sigma_2 t_1 t_2 + \sigma_2^2 t_2^2)\right]. \end{aligned} \tag{56}$$

In the singular case that

$$\begin{vmatrix} \sigma_{11} & \sigma_{12} \\ \sigma_{21} & \sigma_{22} \end{vmatrix} = 0 \tag{57}$$

(Kenney and Keeping 1951, p. 94), it follows that

$$\sigma_{11} \sigma_{22} = \sigma_{12} \sigma_{21} \tag{58}$$

$$y_1 = \mu_1 + \sigma_{11} x_1 + \sigma_{12} x_2 \tag{59}$$

$$y_2 = \mu_2 + \frac{\sigma_{12} \sigma_{21}}{\sigma_{11}} x_2 \tag{60}$$

$$= \mu_2 + \frac{\sigma_{11} \sigma_{21} x_1 + \sigma_{12} \sigma_{21} x_2}{\sigma_{11}} \tag{61}$$

$$= \mu_2 + \frac{\sigma_{21}}{\sigma_{11}} (\sigma_{11} x_1 + \sigma_{12} x_2), \tag{62}$$

so

$$y_1 = \mu_1 + x_3 \tag{63}$$

$$y_2 = \mu_2 + \frac{\sigma_{21}}{\sigma_{11}} x_3, \tag{64}$$

where

$$x_3 = y_1 - \mu_1 \tag{65}$$

$$= \frac{\sigma_{11}}{\sigma_{21}} (y_2 - \mu_2). \tag{66}$$

The standardized bivariate normal distribution takes $\sigma_1 = \sigma_2 = 1$ and $\mu_1 = \mu_2 = 0$. The quadrant probability in this special case is then given analytically by

$$P(x_1 \leq 0, x_2 \leq 0) = P(x_1 \geq 0, x_2 \geq 0) \tag{67}$$

$$= \int_{-\infty}^0 \int_{-\infty}^0 P(x_1, x_2) d x_1 d x_2 \tag{68}$$

$$= \frac{1}{4} + \frac{\sin^{-1} \rho}{2 \pi} \tag{69}$$

(Rose and Smith 1996; Stuart and Ord 1998; Rose and Smith 2002, p. 231). Similarly,

$$P(x_1 \leq 0, x_2 \geq 0) = P(x_1 \geq 0, x_2 \leq 0) \tag{70}$$

$$= \int_{-\infty}^0 \int_0^{\infty} P(x_1, x_2) d x_1 d x_2 \tag{71}$$

$$= \frac{\cos^{-1} \rho}{2 \pi}. \tag{72}$$

SEE ALSO

Box-Muller Transformation, Correlation Coefficient–Bivariate Normal Distribution, Multivariate Normal Distribution, Normal Distribution, Price's Theorem, Trivariate Normal Distribution

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